

DIGITAL IMAGE PROCESSING

LECTURE # 11

IMAGE COMPRESSION-I

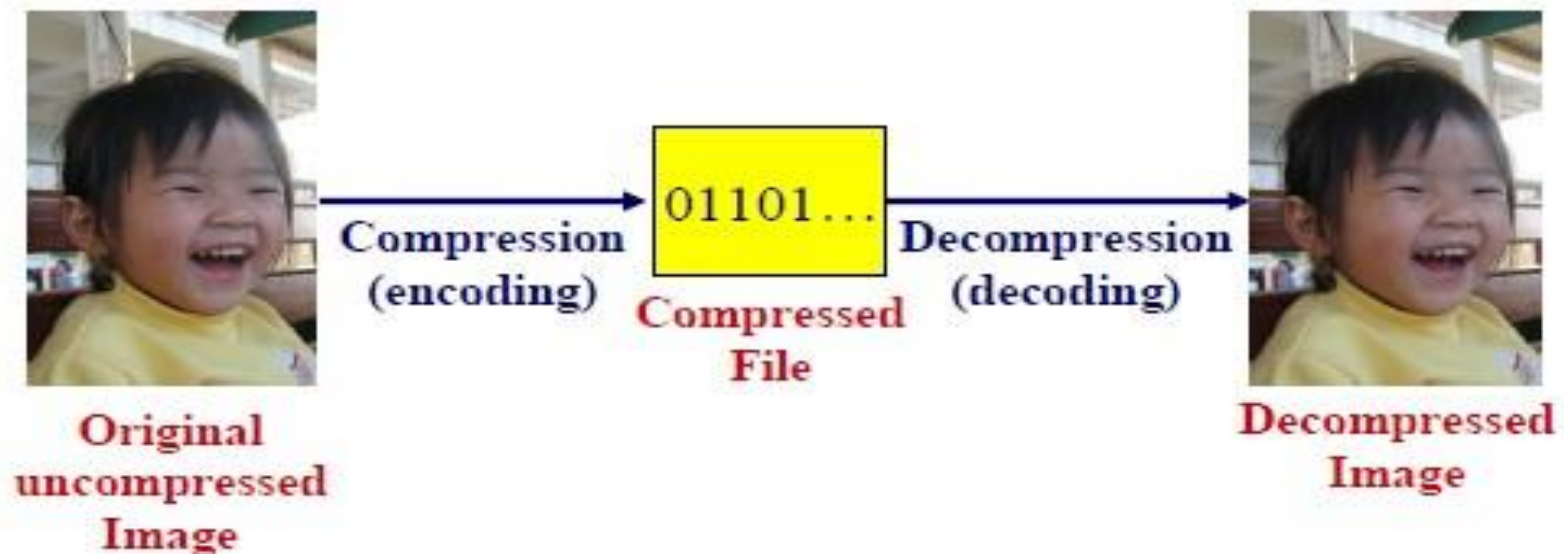
Dr. M. Bilal

Topics to Cover

- ❑ Semester Project Discussion
- ❑ Fundamentals of Image Compression
- ❑ Background
- ❑ Overview
- ❑ Data VS Information
- ❑ Data Redundancy
- ❑ Coding Redundancy
- ❑ Interpixel Redundancy
- ❑ Psychovisual Redundancy
- ❑ Fidelity Criteria
- ❑ Image Compression Model

Fundamentals of Image Compression

- ❑ Digital image contains a lot of data — far too much for practical applications (e.g. problem of storage and efficient transmission)
- ❑ Any data may be represented in different ways, image compression exploits redundancy to achieve reduction in the actual amount of data without (seriously) affecting the quality of the image
- ❑ **Image Compression**- reduce the size of image data file while retaining necessary information



Background

- **Principal objective:**

To minimize the number of bits required to represent an image.

- **Applications**

- ✓ **Transmission:**

Broadcast TV via satellite, teleconferencing etc.

- ✓ **Storage:**

Educational and business documents, medical images (CT, MRI and digital radiology), satellite images, weather maps, geological surveys, e.t.c)

Overview

Image data compression methods fall into two common categories:

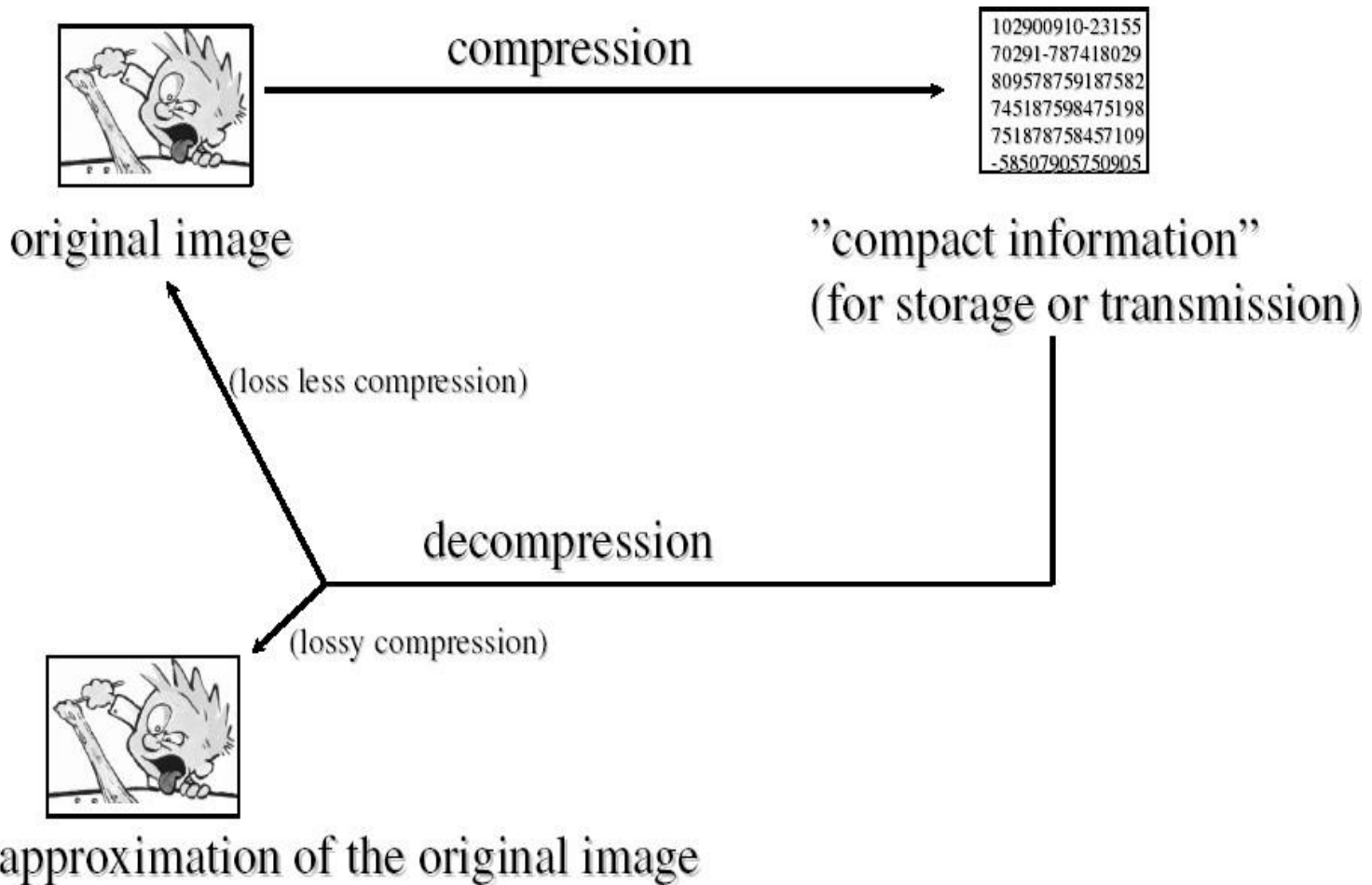
I. Information preserving compression

- Compress and decompress images without losing information
- Compression ratio – 2:1 to 3:1
- Especial for image archiving (storage of legal or medical records)

II. Lossy image compression

- Result in a less than perfect reproduction of the original image
- Provide higher levels of data reduction
- Compression ratio – 10:1 to 20:1 or more
- Applications: –*broadcast television, videoconferencing*

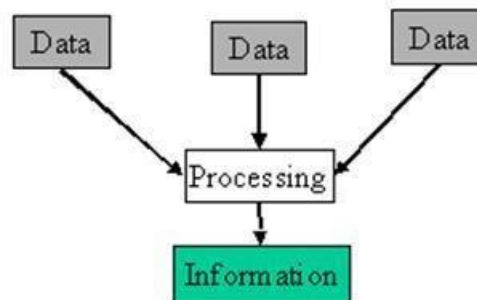
Overview



Data VS Information

- ❑ *Data* is not the same thing as *information*
- ❑ **Data** are the means to convey **information**; various amounts of **data** may be used to represent the same amount of **information**
- ❑ Data that provide no relevant information=**redundant data** or **redundancy**.
- ❑ Image compression techniques can be designed by reducing or eliminating the **Data Redundancy**

Information is created from data



Data VS Information

- Compression ratio – the ratio of the original, uncompressed image file and the compressed file

$$\text{Compression Ratio} = \frac{\text{Uncompressed File Size}}{\text{Compressed File Size}} = \frac{\text{SIZE}_U}{\text{SIZE}_C}$$

- Compression ratio = $100/10 = 10$
- Compression ratio of 10 here indicates that 90% of the data in the uncompressed file is redundant

Data Redundancy

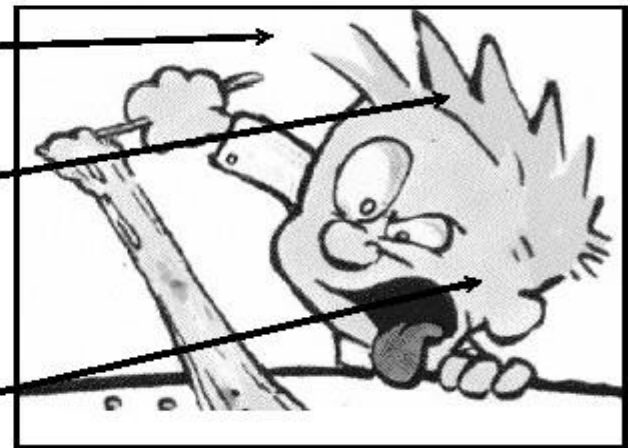
Three basic data redundancies

- Coding Redundancy
- Interpixel Redundancy
- Psychovisual Redundancy

CR: some graylevels are more common than others

IR: the same graylevel covers large areas

PVR: the eye can only resolve 32 graylevels locally

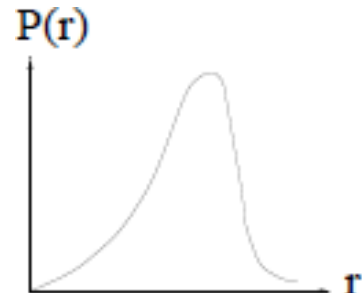


Coding Redundancy

- ❑ A natural m-bit coding method assigns m-bit to each gray level without considering the probability that gray level occurs with: **Very likely to contain coding redundancy**
- ❑ **Basic concept:**
- ❑ Utilize the probability of occurrence of each gray level (histogram) to determine length of code representing that particular gray level: variable-length coding.
- ❑ Assign shorter code words to the gray levels that occur most frequently or vice versa.

Coding Redundancy

- If the gray level of an image is coded in a way that uses more code words than necessary to represent each gray level, then the resulting image is said to contain coding redundancy
- Let $P_r(r_k) = n_k / n$, (histogram) in normal 8-bit representation, each gray level r_k is denoted by a code of 8-bit, L = average number of bit per pixel = 8
- If the histogram is **not** flat then it means some gray levels occur more often than others, L can be reduced by using *variable* length codes l_k for gray level r_k
- $L = \sum_{k=0}^{L-1} P_r(r_k) \cdot l_k < 8$ if $l_k < 8$ for the more probable gray levels r_k



Coding Redundancy

Let $0 \leq r_k \leq 1$: gray levels (discrete random variable)

$p_r(r_k)$: probability of occurrence of r_k

n_k : number of pixels that r_k appears in the image

n : total number of pixels in an image

L : number of gray levels

$l(r_k)$: number of bits used to represent r_k

L_{avg} : average length of code words assigned to the grey levels

$$L_{avg} = \sum_{k=0}^{L-1} l(r_k) p_r(r_k) \quad \text{where} \quad p_r(r_k) = \frac{n_k}{n}, \quad k = 0, 1, \dots, L-1$$

Hence, total number of bits required to code an $M \times N$ image is MNL_{avg}

For a natural m -bit coding $L_{avg} = m$.

Coding Redundancy

Let us assume an 8-level image

8-levels: can be represented by 3-bits, $m = 3$.

Normalized histogram of the image is given by:

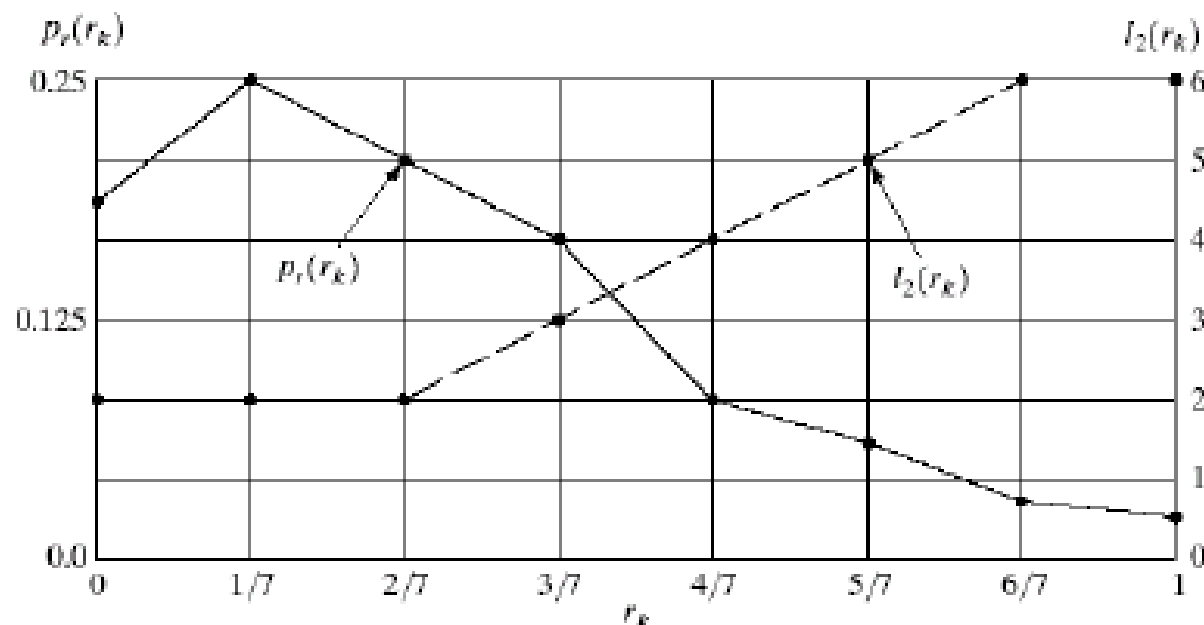


FIGURE 8.1

Graphic representation of the fundamental basis of data compression through variable-length coding.

Coding Redundancy

r_k	$p_r(r_k)$	Code 1	$l_1(r_k)$	Code 2	$l_2(r_k)$
$r_0 = 0$	0.19	000	3	11	2
$r_1 = 1/7$	0.25	001	3	01	2
$r_2 = 2/7$	0.21	010	3	10	2
$r_3 = 3/7$	0.16	011	3	001	3
$r_4 = 4/7$	0.08	100	3	0001	4
$r_5 = 5/7$	0.06	101	3	00001	5
$r_6 = 6/7$	0.03	110	3	000001	6
$r_7 = 1$	0.02	111	3	000000	6

TABLE 8.1
Example of
variable-length
coding.

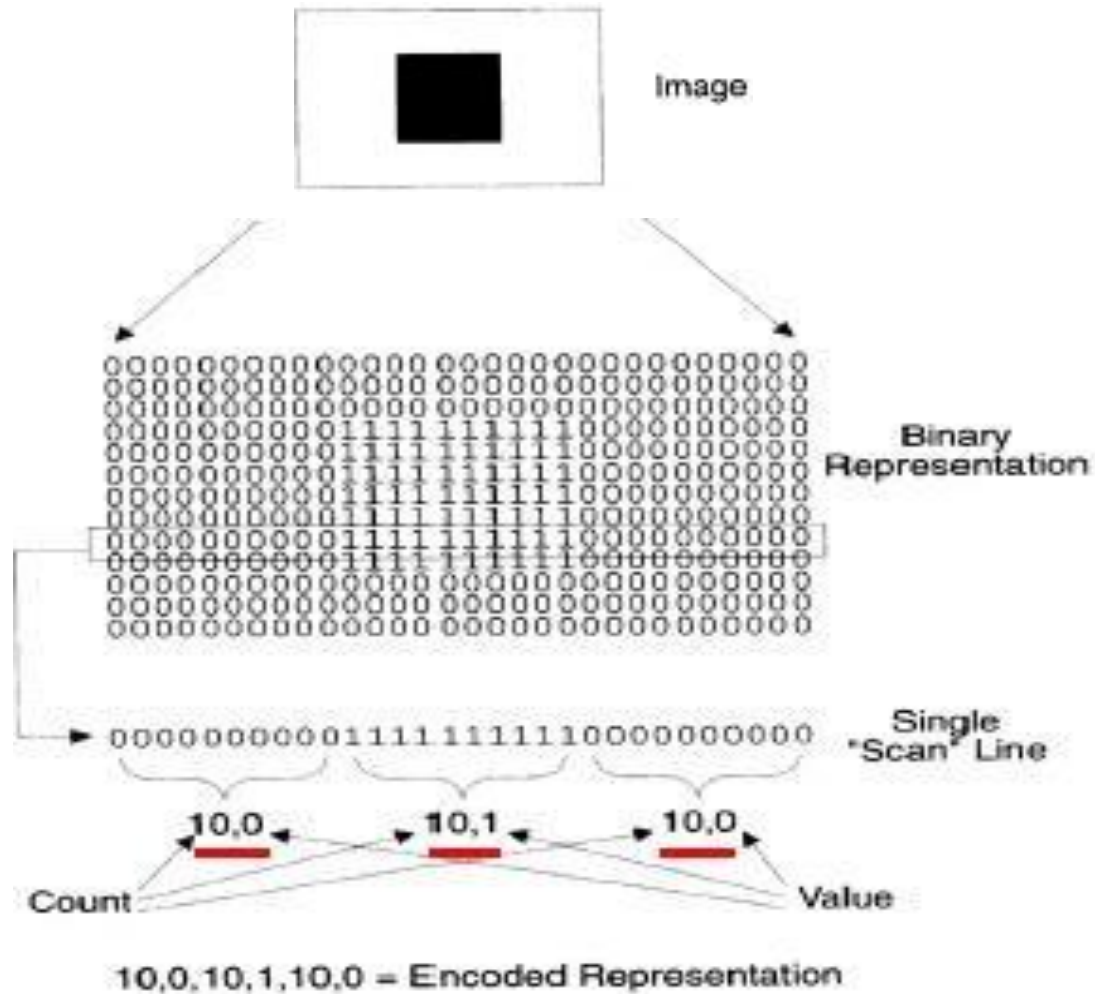
$$L_{avg_1} = m = 3 \text{ bits, and}$$

$$\begin{aligned}
 L_{avg_2} &= \sum_{k=0}^7 l_2(r_k) p_r(r_k) \\
 &= 2(0.19) + 2(0.25) + 2(0.31) + 3(0.16) \\
 &\quad + 4(0.08) + 5(0.06) + 6(0.03) + 6(0.02) = 2.7 \text{ bits}
 \end{aligned}$$

Interpixel Redundancy

- ❑ Caused by High Interpixel Correlations within an image, i.e., gray level of any given pixel can be reasonably predicted from the value of its neighbors (information carried by individual pixels is relatively small) spatial redundancy, geometric redundancy, interframe redundancy (in general, interpixel redundancy)
- ❑ To reduce the interpixel redundancy, mapping is used. The mapping scheme can be selected according to the properties of redundancy.
- ❑ An example of mapping can be to map pixels of an image:
 $f(x,y)$ to a sequence of pairs: $(g_1, r_1), (g_2, r_2), \dots, (g_i, r_i), \dots$
 g_i : i th gray level r_i : run length of the i th run
- ❑ Removing interpixel redundancy is lossless

Interpixel Redundancy



Interpixel Redundancy

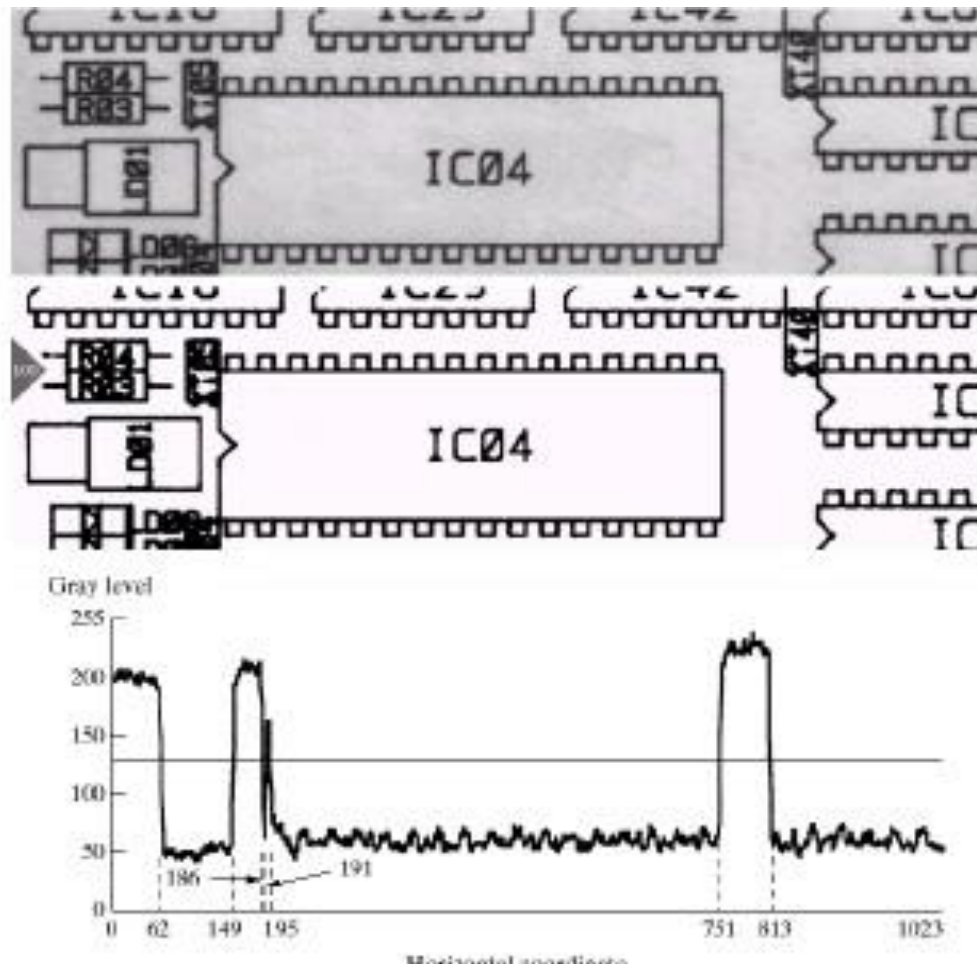
An image with size
 $1024(y) \times 343(x)$

Consider the 1024 pixels
 along $x=100$ (horizontal
 direction)

To represent the binary
 image we only need 8 pairs
 (8 runs) because the gray
 level changes 7 times

Line 100:

(1,63) (0,87) (1,37) (0,5)
 (1,4) (0,556) (1,62) (0,210)



Psychovisual Redundancy

- ❑ The eye does not respond with equal sensitivity to all visual information.
- ❑ Certain information has less relative importance than other information in normal visual processing **psychovisually redundant** (which can be eliminated without significantly impairing the quality of image perception).
- ❑ The elimination of psychovisually redundant data results in a loss of quantitative information **lossy data compression method**.
- ❑ Image compression methods based on the elimination of psychovisually redundant data (usually called **quantization**) are usually applied to commercial broadcast TV and similar applications for human visualization.

Psychovisual Redundancy

If the image will only be used for visual observation (i.e. illustrations on the web etc), a lot of the information is usually psycho-visually redundant. It can be removed without changing the visual quality of the image. This kind of compression is usually irreversible.



0.5kB



0.05kB

Psychovisual Redundancy



Original image
(256 levels)



Normal quantization
(16 levels)



IGS quantization
(16 levels)

Psychovisual Redundancy

- ❑ Improved Grayscale (IGS) Quantization
- ❑ If we use fewer bits/graylevels to represent an image it causes the false contouring. To reduce this effect, a level of randomness is added to the artificial edges associated with false contouring by IGS quantization.

Psychovisual Redundancy

IGS quantization procedure:

1. Set initial SUM = 0000 0000
2. if most significant 4 bits of current pixel A = 1111
 $\text{new_SUM} = B + 0000$

else

$\text{new_SUM} = B + \text{least significant 4 bits of old SUM}$

<i>Pixel</i>	<i>Gray Level</i>	<i>Sum</i>	<i>IGS Code</i>
$i - 1$	A N/A B	0000 0000	N/A
i	0110 1100	0110 1100	0110
$i + 1$	1000 1011	1001 0111	1001
$i + 2$	1000 0111	1000 1110	1000
$i + 3$	1111 0100	1111 0100	1111

Fidelity Criteria

27

Quantify the nature and extent of information loss.

Objective fidelity criteria:

Level of information loss can be expressed as a function of the original (input) and compressed-decompressed (output) image.

Given an $M \times N$ image $f(x, y)$ (original image), its compressed-then-decompressed image: $\hat{f}(x, y)$, then the error between corresponding values are given as:

$$e(x, y) = \hat{f}(x, y) - f(x, y)$$

Total error is given by:

$$\sum_{x=0}^{M-1} \sum_{y=0}^{N-1} [\hat{f}(x, y) - f(x, y)]$$

Fidelity Criteria

28

Subjective fidelity criteria:

-Let a number of test persons grade the images as bad/OK/good etc.

Since most decompressed images ultimately are viewed by human beings, measuring image quality by the subjective evaluations of a human observer often is more appropriate.

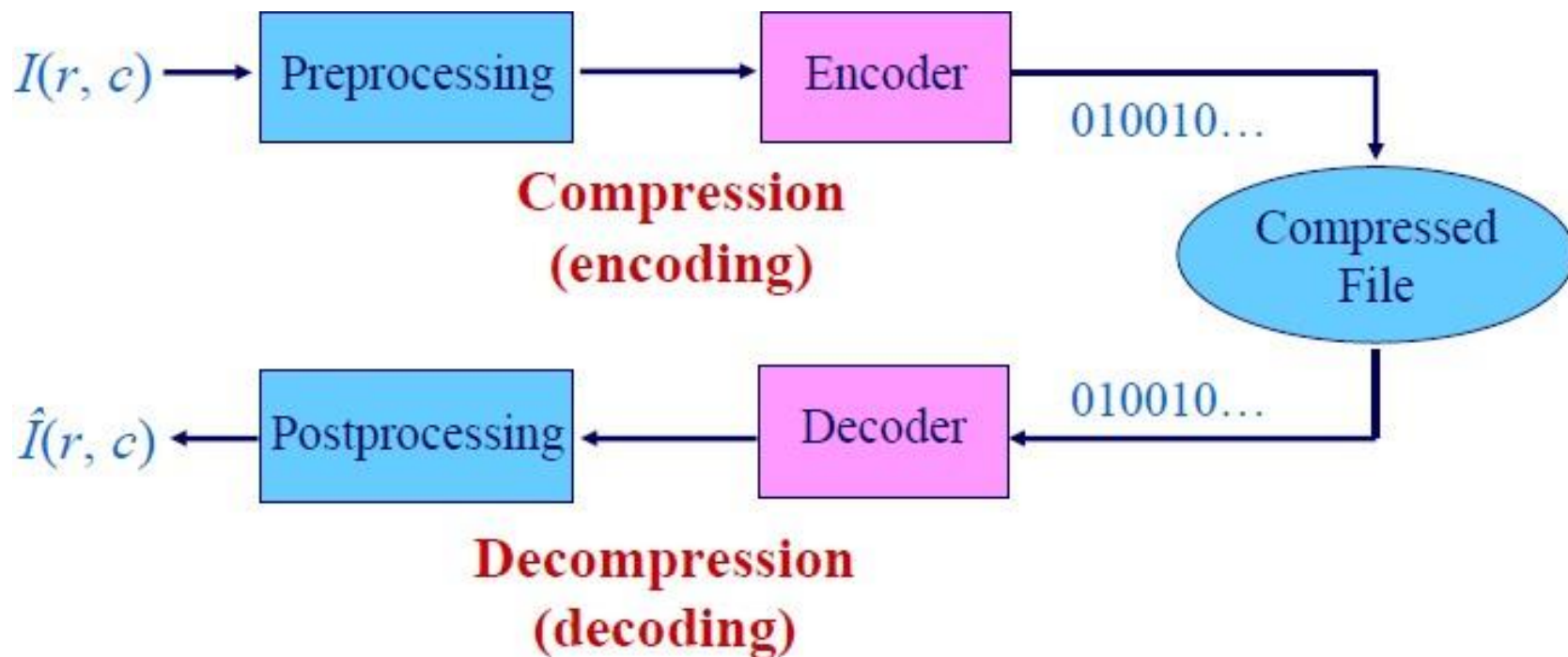
Method: evaluations are made using an absolute rating scale, by means of side-by-side comparisons of $f(x,y)$ and $\hat{f}(x,y)$.

TABLE 8.3
Rating scale of the
Television
Allocations Study
Organization.
(Freundtall and
Behrend.)

Value	Rating	Description
1	Excellent	An image of extremely high quality, as good as you could desire.
2	Fine	An image of high quality, providing enjoyable viewing. Interference is not objectionable.
3	Passable	An image of acceptable quality. Interference is not objectionable.
4	Marginal	An image of poor quality; you wish you could improve it. Interference is somewhat objectionable.
5	Inferior	A very poor image, but you could watch it. Objectionable interference is definitely present.
6	Unusable	An image so bad that you could not watch it.

Image Compression Model

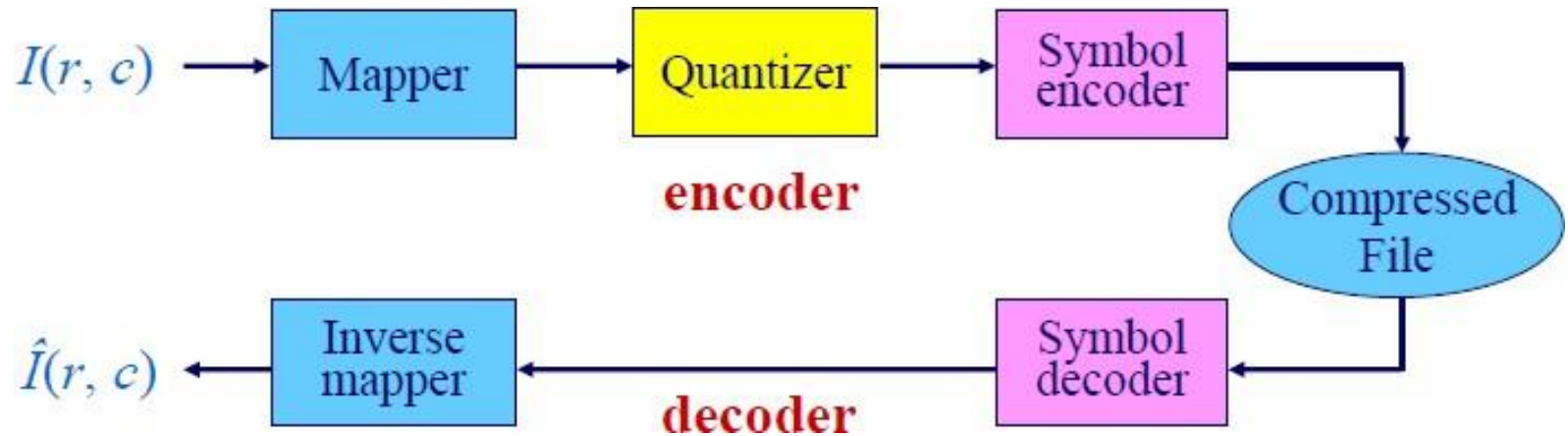
29



- ❑ The encoder removes input redundancies (interpixel, psychovisual, and coding redundancy)

Image Compression Model

30



- mapper - reduce interpixel redundancy, reversible (run-length coding, Transform coding etc.)
- quantizer - reduce Psychovisual redundancy, irreversible (uniform quantization, nonuniform quantization)
- symbol encoder - reduce coding redundancy, reversible (Huffman coding, LZW etc.)

References

31

1. DIP by Gonzalez

For any query Feel Free to ask



32

